

Matteo Cacioppo

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ML Researcher — Agentic AI Systems & Deep Learning

ML researcher with a physics background, working on deep learning for biomedical signal processing and on production-oriented agentic AI systems: multi-agent orchestration, retrieval over heterogeneous technical data, evaluation, and tracing.

Experience

Research Fellow — ENEA Frascati

09/2025 – Present

Horizon Europe project NARCOSIS — multi-agent AI platform for forensic identification of new psychoactive substances (NPS)

- Contributed to the development of a full-stack agentic platform integrating LLM-based orchestration with spectral classification (Raman, FTIR, SWIR) for substance identification from analytical data.
- Built a function-routing orchestrator (ReAct-style loop) that dynamically selects retrieval, spectral-analysis, and reasoning tools across multi-step queries.
- Designed an agentic RAG system combining knowledge-graph retrieval and vector search over heterogeneous technical corpora (papers, structured reports, spectral data).
- Integrated MCP (Model Context Protocol) servers to modularise compute, chemistry, vision, and spectral-analysis capabilities; implemented access control and credential handling for tool servers.
- Instrumented agent execution with structured, step-level trace logging for debugging, failure analysis, and explainability.
- Defined evaluation protocols for agentic outputs: retrieval relevance, tool-selection correctness, and end-to-end task success on curated query sets.
- Profiled local LLM inference for latency, throughput, and context budget; tuned serving and model choices against response-time constraints.
- Applied compliance-by-design principles (EU AI Act, NIS2) through local model execution and auditable agent traces.

Publications

Real-Time Motion Correction in Magnetic Resonance Spectroscopy: a complete overview of the reconstruction process

10/2025

Poster, XIV Workshop of the International School on Magnetic Resonance and Brain Function (ISMRFB), Erice, Italy

Real-Time Motion Correction in Magnetic Resonance Spectroscopy: AI solution inspired by fundamental science

06/2025

arXiv:2509.24676 (2025). Poster, EuCAIFCon, Cagliari, Italy

Education

MSc in Physics — 110/110 cum laude — Sapienza University of Rome

10/2022 – 01/2025

Thesis: LSTM Neural Network for Real-Time Motion Artefact Correction in MRS Scans — Advisor: Prof. S. Giagu

BSc in Physics — Sapienza University of Rome

10/2015 – 11/2022

Thesis: Raman spectroscopy in the frequency and Fourier-transform domains — Advisor: Prof. T. Scopigno

Selected Projects

Real-Time Artefact Correction in MRS — RECENTRE project

2025

- Implemented an LSTM network predicting patient motion from navigator data, applying affine transformation parameters to correct spatial misalignment in MRS signals.
- Designed a real-time pipeline and evaluated it on patient data for noise reduction and artefact correction.

LIS Gesture Classification — Multimodal deep learning

2025

- Built an LSTM classifier for 126 Italian Sign Language gestures, fusing RGB and radar features from pre-trained and custom CNNs via late fusion.
- Handled variable-length sequences with padding and packed sequences; compared fusion strategies on the official test set.

Graph Coloring with Graph Neural Networks — Physics-inspired ML

2023

- Implemented a GNN (DGL) for graph coloring, with per-node colour distributions via softmax and mean-aggregation message passing.
- Defined a physics-inspired loss based on the Potts-model Hamiltonian, with an auxiliary conflict-count metric at inference.

Skills

Programming: Python, PyTorch (advanced); C, C++, $\text{\LaTeX}$ (basic)

AI systems: Multi-agent orchestration, RAG, MCP, tool-calling, local LLM serving, evaluation and tracing

Tools: Docker, Git, GitHub, Linux, React

Languages: Italian (native), English (professional working proficiency)